Regression Assignment – Insurance Charges Prediction

1.Problem Statement	2 .Solution
To predict medical insurance charges based on age, gender, BMI, number of children, and smoking habits	I used regression models and compared their performance using R^2 Score and Mean Absolute Error (MAE) to find the best model.

3. Data Preprocessing

- Checked for missing values.
- Removed one duplicate record.
- Converted categorical columns into numerical values using One-Hot Encoding.
- Split the dataset into 80% training and 20% testing .

4. Algorithms I Used

- Linear Regression
- Decision Tree Regression
- Random Forest Regression
- Gradient Boosting Regression
- Extra Trees Regression

5. Model Comparison

Algorithm	R^2 Score
Decision Tree	0.7831
Extra Trees	0.8456
Gradient Boosting	0.8999
Linear Regression	0.8046
Random Forest	0.8747

6. Why I Chose Gradient Boosting

I selected Gradient Boosting Regression because it gave the highest R^2 score of 0.8999 and the lowest MAE of 2525.85 among the five models.